

on the pan-tilt mechanism, allowing the camera to follow the servo movement.

Fig. 1 shows the authors' prototype and the real-time testing setup for the 50+ FPS face-tracking system.

The main controller used in this system is the ESP32, which receives commands from the laptop over the ESPconnect and controls the servo motors accordingly. A VL53L0X time-of-flight (ToF) distance sensor continuously measures the distance to the detected object and sends the readings back to the laptop, where they are overlaid on the live video feed. The complete list of components required to build this system is provided in Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the 50+ FPS face-tracking system using the laptop-ESP32 duo. The

system is built around the ESP32, which controls the pan-tilt mechanism, reads the VL53L0X ToF sensor, and updates the OLED display. A 7V-9V DC supply passes through a reverse-polarity protection diode and a 5V regulator. The regulated 5V rail powers the PCA9685 PWM driver and the two servos, while a 100µF capacitor helps minimise voltage fluctuations caused by servo movement. The ESP32, OLED, and VL53L0X are powered from their appropriate regulated supply pins, with all grounds connected to a common reference.

The ESP32 communicates with the PCA9685, VL53L0X, and OLED through the I²C bus, typically using GPIO 21 (SDA) and GPIO 15 (SCL). The PCA9685 receives position commands from the ESP32 and generates PWM signals for the pan and tilt servos through its first two channels. The pan servo provides horizontal movement, while the tilt servo provides vertical movement. The remaining 14 channels are available for future expansion. The VL53L0X measures the distance to the object and sends the data to the ESP32, while the OLED displays information such as distance and servo status.

A USB webcam is mounted on the pan-tilt mechanism and connected to the laptop. The laptop runs the YOLOE face-detection software, which determines the face position in the camera frame and sends corresponding

movement commands to the ESP32. The ESP32 processes these commands and drives the servos through the PCA9685 to keep the detected face near the centre of the frame.

Thus, the laptop performs face detection, while the ESP32 handles servo control, distance measurement, and display functions. This division of tasks enables fast and responsive face tracking. The PCA9685's additional channels also allow the system to be extended for multi-axis robotic applications.

Fig. 3 shows the pan-tilt camera mount used to provide two-axis movement to the camera. A USB camera is mounted on the top of the mechanism and can be rotated horizontally (pan) and vertically (tilt) using two servo motors. The servos are mounted in the black mechanical frame, providing controlled movement of the camera in both directions.

The camera is connected to the laptop through the USB cable, while the servo motors are controlled by the ESP32. The wooden base provides a stable platform for the assembly. This arrangement allows the camera to continuously change its viewing direction and track a detected face or object by adjusting the pan and tilt angles.

Software/coding

YOLOE represents a significant evolution of traditional YOLO models. While conventional YOLO detectors are limited to a fixed set of predefined object classes, YOLOE uses a prompt-driven mechanism that allows users to define objects of interest dynamically without retraining. In conventional workflows, detecting a new object requires collecting a suitable dataset and performing computationally expensive training or fine-tuning, often using a high-end GPU. YOLOE overcomes this limitation through



Fig. 3: Pan-tilt camera mount for two-axis movement

```
from ultralytics import YOLOE
model = YOLOE("yoloe-11s-seg.pt") # use 11l for larger model
names = ["person", "remote", "glasses", "magnifying glass", "smartphone", "clock", "banana", "book"]
model.set_classes(names, model.get_text_pe(names))
model.export(format="ncnn", imgsz=640)
```

Fig. 4: Conversion Python script